Srija Baratam

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.49
FIITJEE Global School	Class XII (CBSE)	2024	93.40%
Maharishi Vidya Mandir, Polachery, Chennai	Class X (CBSE)	2022	92.20%

POSITIONS OF RESPONSIBILITY

Astro-Tech Core Member, Cepheid:

Developed practical expertise in telescope operation, alignment, calibration, and astronomical observations while supporting club events and observation sessions.

PROJECTS

2D Tower Defense Game Engine: Tower_Defense

- Built a custom 2D game engine from scratch in C++17/SDL2 to demonstrate measurable performance trade-offs in real-time collision detection, without relying on any game engine frameworks.
- Implemented a spatial hash grid (O(1) avg lookup) and BFS pathfinding with dynamic path invalidation, then extended the project with a FastAPI/SQLite backend, React leaderboard, and pandas/matplotlib analytics pipeline.
- Delivered a full working pipeline from live gameplay to a ranked, win/loss-aware leaderboard, with session and benchmark data automatically visualized through pandas/matplotlib for measured performance analysis.

Scooter Telemetry Analysis: Scooter_Telemetry_Analysis

- Built a 5-stage IoT data pipeline (ingestion, sessionization, KPI computation, anomaly detection, and report generation) to process raw electric scooter telemetry.
- Achieved 99% validation pass rate across 1,500 samples from 3 vehicles with rule-based anomaly detection for temperature spikes, SOC glitches, and GPS jumps.
- Generated per-vehicle JSON summaries and PNG visualizations, surfacing 208 anomalies including overheating patterns invisible in raw data.

Detection of Nanohertz Gravitational Waves using Pulsar Timing Arrays: Detection_of_nHz_GW_using_PTA

- Simulated a 10-year pulsar timing array to model gravitational-wave signatures from supermassive black hole binaries using spectral analysis and Cholesky-based correlation modelling.
- Reproduced the Hellings-Downs angular correlation and f^(-13/3) power spectrum, validating the simulation against theoretical predictions used in real PTA experiments.

SKILLS

Technical Skills:

Programming Languages: C, C++, Python, JavaScript, SQL, MATLAB

Backend & Systems: SDL2, FastAPI, Pydantic, SQLite

Frontend: React, Vite, HTML, CSS

Data & Analytics: pandas, NumPy, Matplotlib

Developer Tools: Git, GitHub, VS Code, Visual Studio

Soft Skills: Effective Communication, Problem Solving, Team Collaboration.

RELEVANT COURSES

Computer Science and Programming:

Introduction to Programming, Data Structures & Applications, Computational Physics.

Artificial Intelligence:

Introduction to Artificial Intelligence, Introduction to Probability, Introduction to Statistics.

Mathematics: Calculus I, Calculus II, Elementary Linear Algebra, Differential Equations, Maths for Physics, Mathematical Physics, Advanced Mathematical Physics.

EXTRACURRICULAR

Assisted in organizing and hosting public **telescope observation sessions** for students and visitors.

Engaged regularly in **chess and badminton** as hobbies, building focus and analytical thinking, and physical fitness through consistent practice.

Volunteered in **NSS**, taking part in service activities and campus events, which improved teamwork and social awareness.